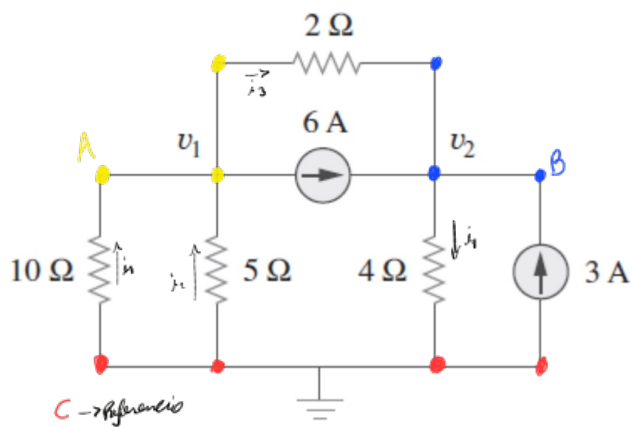


3.2 Para o circuito da Figura 3.51, obtenha v_1 e v_2 .



$$i_1 = \frac{-V_A}{10} \quad i_2 = \frac{-V_A}{5} \quad i_3 = \frac{V_A - V_B}{2} \quad i_4 = \frac{V_B}{4}$$

$$\text{* No } \emptyset: 3 + 6 + i_3 = i_4 \quad \text{* No } A: i_2 + i_3 = 6 + i_4$$

$$\frac{V_B}{4} = 9 + \frac{(V_A - V_B)}{2}$$

$$\frac{-V_A}{10} - \frac{V_A}{5} = 6 + \frac{(V_A - V_B)}{2}$$

$$V_B = 36 + 2V_A - 2V_B$$

$$-V_A - 5V_A = 60 + 5V_A - 5V_B$$

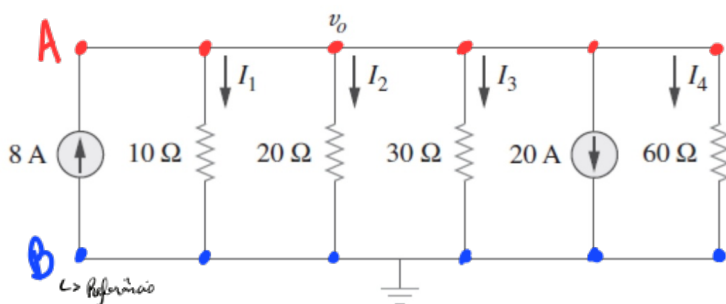
$$3V_B - 2V_A = 36$$

$$5V_B - 11V_A = 60$$

$$\begin{cases} 5V_B - 11V_A = 60 \\ 3V_B - 2V_A = 36 \end{cases}$$

$$\begin{cases} V_B = 12 \\ V_A = 0 \end{cases}$$

3.3 Determine as correntes I_1 a I_4 e a tensão v_o no circuito da Figura 3.52.



$$i = \frac{(V_{\text{potencial}} - V_{\text{potencial}})}{R}$$

$$V_B = 0 \rightarrow \text{Referência}$$

$$i_1 = \frac{V_A}{10} \quad i_2 = \frac{V_A}{20} \quad i_3 = \frac{V_A}{30} \quad i_4 = \frac{V_A}{60}$$

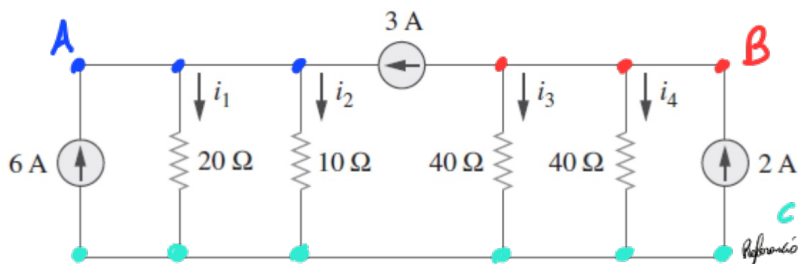
$$\text{* No } A: i_1 + i_2 + i_3 + i_4 = 0$$

$$\frac{V_A}{10} + \frac{V_A}{20} + \frac{V_A}{30} + \frac{V_A}{60} + 20 = 0$$

$$6V_A + 3V_A + 2V_A + V_A = -720$$

$$12V_A = -720 \rightarrow V_A = -60V$$

3.4 Dado o circuito da Figura 3.53, calcule as correntes i_1 a i_4 .



$$i_1 = \frac{V_A}{20} \quad i_2 = \frac{V_A}{10} \quad i_3 = \frac{V_B}{40} \quad i_4 = \frac{V_B}{40}$$

$$\text{* No } A: i_1 + i_2 = 9$$

$$\text{* No } B:$$

$$\frac{V_A}{20} + \frac{V_A}{10} = 9$$

$$3 + \frac{V_B}{40} + \frac{V_B}{40} = 2$$

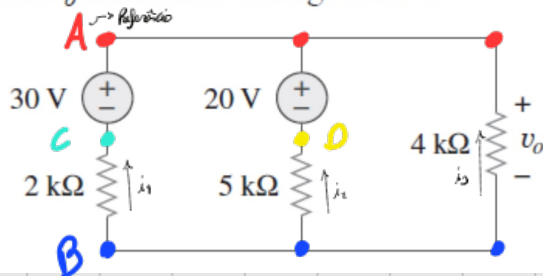
$$3V_A = 180$$

$$2V_B = -40$$

$$V_A = 60V$$

$$V_B = -20V$$

3.5 Obtenha v_o no circuito da Figura 3.54.



$$V_C = -30 \quad V_D = -20$$

$$i_1 = \frac{V_B - V_C}{2000} \quad i_2 = \frac{V_B - V_D}{5000} \quad i_3 = \frac{V_B}{4000}$$

$$\text{* No } \emptyset: i_1 + i_2 + i_3 = 0$$

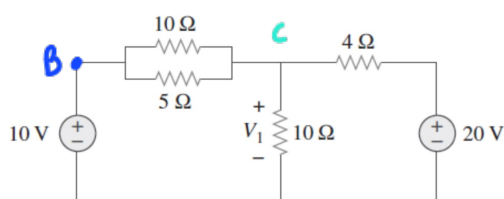
$$\frac{V_B + 30}{2000} + \frac{V_B + 20}{5000} + \frac{V_B}{4000} = 0$$

$$10V_B + 300 + 4V_B + 80 + 5V_B = 0$$

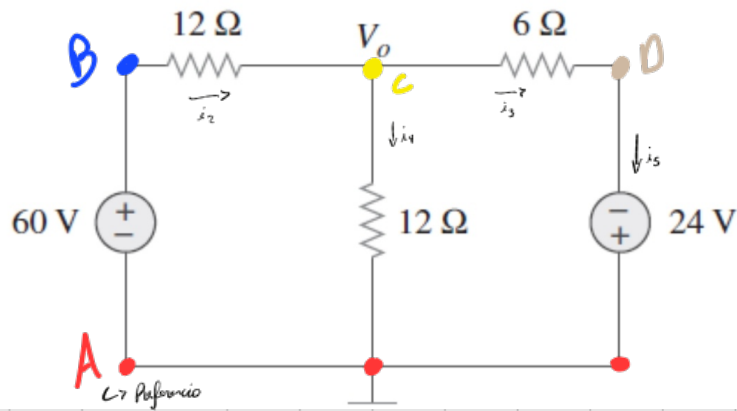
$$19V_B = -380$$

$$V_B = -20V$$

3.6 Use a análise nodal para calcular V_1 no circuito da Figura 3.55.



3.11 Determine V_o e a potência dissipada em todos os resistores no circuito da Figura 3.60.



$$\begin{aligned}
 V_D &= -60V & V_D &= 24V \\
 i_4 &= \frac{V_C}{12} & i_2 &= \frac{V_D - V_C}{12} & i_5 &= \frac{V_C - V_D}{6} \\
 \text{KCL: } i_2 &= i_4 + i_5 \\
 \frac{V_D - V_C}{12} &= \frac{V_C}{12} + \frac{V_C - V_D}{6} \\
 V_D - V_C &= V_C + 2V_C - 2V_D \\
 V_D - 4V_C + 2V_D &= 0 \\
 -60 - 4V_C + 48 &= 0 \\
 \boxed{V_C} &= -3V
 \end{aligned}$$